

Demonstrating Xstrings: 3D Printing Cable-driven Mechanism for Actuation, Deformation, and Manipulation

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Abstract

In this Demo, we present Xstrings, a method for designing and fabricating 3D printed objects with integrated cable-driven mechanisms that can be printed in one go without the need for manual assembly. Xstrings supports four types of cable-driven interactions—bend, coil, screw and compress—which are activated by applying an input force to the cables. To facilitate the design of Xstrings objects, we developed a design tool that allows users to embed cable-driven mechanisms into the object geometry based on the desired interaction by automatically placing joints and cables at the respective locations. The application potential of Xstrings is demonstrated through examples such as manipulable gripping, bionic robot manufacturing, and dynamic prototyping.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Systems and tools for interaction design**.

Keywords

Cable-driven Mechanism, Personal Fabrication, 3D Printing

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1 Introduction

Prior HCI research has focused on integrating mechanisms into physical objects to create dynamic, articulated, and shape-changing

structures. Researchers have explored mechanisms such as compliant [5, 7, 18], elastic [3, 12, 15], bi-stable [10, 19], and shape-memory [8, 11, 16, 17] designs, enabling users to customize dynamic devices and tangible interfaces.

Cable-driven mechanisms are central to modern robotics, offering precision, adaptability, and flexibility. Applications range from robotic actuation [13] to intricate control systems [14]. Nikafrooz et al. [9] designed a robotic hand with cable-driven fingers mimicking tendons, while Colan et al. [2] applied similar principles to advanced surgical tools.

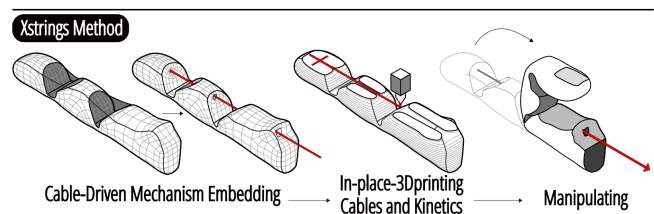


Figure 1: Embedding cable-driven mechanism with the Xstrings workflow.

Despite their advantages, cable-driven mechanisms present challenges for less experienced 3D-printing users. Designing specific motions requires precise cable positioning and tension adjustment. Optimization-based methods, such as those by Megaro et al. [6] and Li et al. [4], have streamlined the inverse design of cable-driven systems, enabling outputs like realistic character animations and actuated grippers. However, the manual fabrication process remains labor-intensive and time-consuming, involving tasks like threading cables through holes and tying precise knots. These challenges limit precision, reproducibility, and scalability in producing consistent cable-driven structures.

To address challenges in designing and fabricating cable-driven mechanisms, we present Xstrings, a method that integrates these mechanisms with 3D object geometries for one-step fabrication using dual-material FDM 3D printing. Xstrings supports four motion primitives—bend, screw, coil, and compress—activated by input force on cables. To simplify the design process, we developed a

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tool that automates cable and joint placement based on desired motions, eliminating lengthy production times and manual labor. Targeting 3D printing users, Xstrings lowers the barrier to entry, fosters creativity, and enhances accessibility. We also evaluate how printing parameters like temperature and speed affect cable quality, including tensile strain and interaction repeatability. Through 3D-printed examples such as dynamic fashion, bionic robotics, and interactive prototypes, Xstrings demonstrates its potential to accelerate fabrication for both novice and expert users. In general, our contribution include:

- A one-step fabrication method for cable-driven mechanisms, integrating cables, joints, and linkages directly into 3D-printed objects without manual assembly.
- An interactive design tool that simplifies embedding cable-driven mechanisms into objects by decimating structures into links and supporting four motion primitives (bend, coil, screw, compress) to approximate desired motions.
- A technical evaluation of how printing parameters (temperature, extrusion, density, and print path) affect the performance and durability of printed cables.
- Examples of 3D-printed applications showcasing Xstrings, including gripping devices, bionic robotics, and dynamic prototypes, demonstrating the method's versatility and accessibility.

2 Xstrings METHOD

To begin with, we identify the key components that need to be included in an Xstrings object, referencing classic cable-driven mechanisms. Typically, there are three primary components used to connect a cable to a mechanical structure: (1) **Anchor Points**, (2) **Threaded Points**, and (3) **Exposed Points**, as illustrated in the example of a cable-driven finger in Fig. 1. Below, we introduce the characteristics of each component and provide a detailed explanation of their corresponding manufacturing techniques in Xstrings.

Anchor Points: Cable-driven mechanisms rely on precise control of the end effector (e.g., the fingertip) using cables. Traditional methods achieve this by tying knots, but since 3D printers cannot tie knots, we propose an alternative: T-shaped anchors. These anchors are securely fixed by the surrounding material, as shown in Fig. 2, eliminating the need for thermal adhesion, which may fail if materials do not fuse successfully [1].

Threaded Points: Threaded points are locations where cables pass through holes in the structure, enabling rectilinear sliding motions. While traditionally created by drilling and threading, achieving smooth motion in layer-by-layer 3D printing poses challenges due to friction and resistance between the cable and printed materials. Our parameter optimization addresses these issues to ensure efficient actuation.

By exploring and combining these different types of joints, we can expand the range of applications for 3D-printed cable-actuated mechanical systems and tailor our printed objects to meet the specific needs of various scenarios.

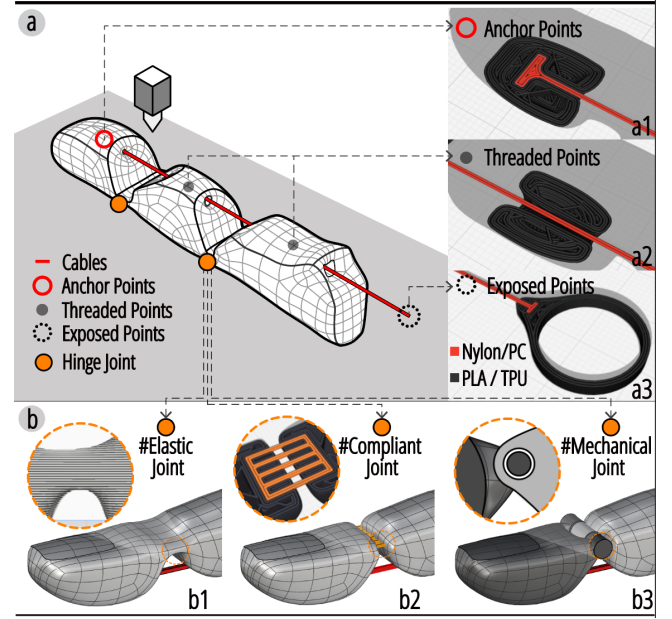


Figure 2: (a) Three types of relationships between cables and mechanical components: Anchor Points, Threaded Points, and Exposed Points. (b) Three types of hinge joints as options: Elastic, Compliant, and Mechanical.

3 Motion Primitives

To automate the creation of cable-driven 3D-printable mechanisms, we focus on their fundamental motions. These include four key one-degree-of-freedom (one-DOF) primitives: Bend, Coil, Screw, and Compress, as shown in Fig. 3. Each represents common motions in cable-driven designs.

We illustrate these primitives using a rectangular cuboid modified with cables, joints, and gaps to enable actuation. These simple building blocks can be combined to create a wide range of motion designs. In the Design Tool section, we demonstrate how they can be customized and combined for user-defined motions.

Bend: The Bend Primitive consists of multiple joints connected by a single driving cable, allowing rotational motion between links. This structure is designed using the parametric design tool Grasshopper. The design process begins by creating a single triangular prism and duplicating it in a linear array. These prisms are then used to cut gaps into the geometry, leaving links connected by flexible bridging joints that enable bending. A cable runs through the structure to transmit force, driving the bending motion.

Coil: Building on the Bend Primitive, the Coil Primitive introduces angled gaps between segments to create a helical motion. This results in a coiling movement that is not aligned with the primary axis. The design involves rotating the cutting elements before creating gaps in the geometry, which forms the coil shape. Cables and joints are added to enable actuation, creating a structure that can coil around its axis while maintaining flexibility.

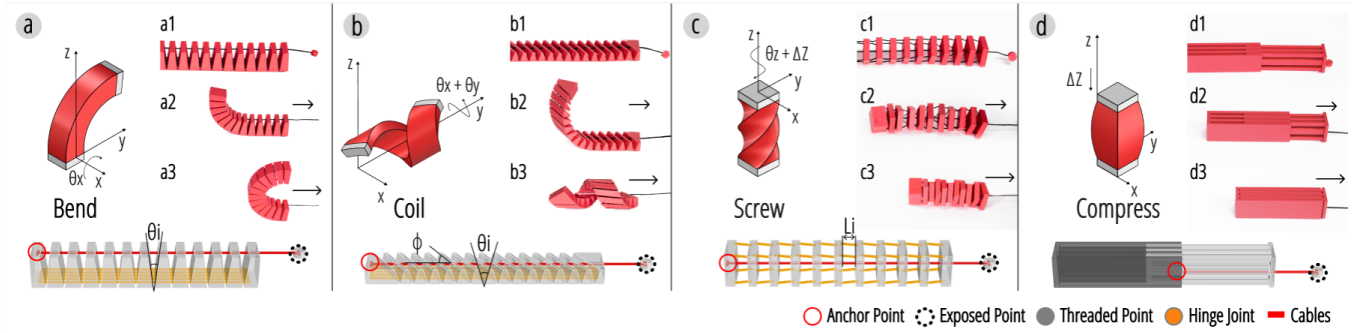


Figure 3: Four kinds of basic motion primitives of Xstrings

Screw: The Screw Primitive is designed as a spring-like structure that twists around its axis while compressing in length. It is constructed by cutting the geometry into thin cuboid links connected by stiffer joints arranged in a circular array. When the cable is actuated, these joints convert the pulling force into a twisting motion, enabling the structure to screw and shorten simultaneously.

Compress: The Compress Primitive features a linear track mechanism with a single degree of freedom, allowing sliding motion along its axis. By pulling the cable, the structure compresses along the track, reducing its length by half. This straightforward mechanism is useful for applications requiring precise linear compression.

4 Design tool

Xstrings software utilizes Rhinoceros 8 as the design environment and Grasshopper as both a computational tool and intermediary interface. The user workflow involves inputting a geometry, selecting desired motion primitives, defining the number of segments and deformation angles, setting a cable curve with adjustable width and end shape, and exporting the final design for printing.

Embedding Primitives: After inputting a geometry (preferably elongated) in Rhinoceros, users can select a primitive from the library (Fig. 4). The interface highlights the current segment in red and others in blue, allowing users to adjust bending degrees and segment counts through sliders. Pre-tested initial values are provided as references. Users can combine multiple primitives by adding segments; for instance, Fig. 4 shows a tentacle with a coil tip and bend primitives along the body.

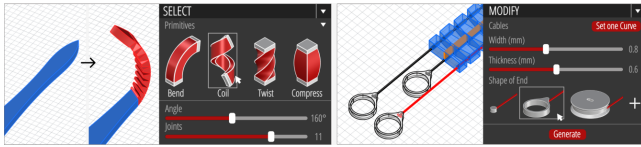


Figure 4: Primitives Embedding process in Xstrings system.

Modifying the Cable: Users can adjust cable parameters, including shape, width, and thickness, based on the estimated force requirements. The Xstrings system automatically generates a T-shaped anchor at the end effector's tip. For threaded points, the tool offsets

the cable's surface by 0.5mm and uses Boolean difference to create holes, maintaining a 0.5mm gap to reduce friction while ensuring precise control. For exposed points, users can choose from three shapes: (1) Dot and (2) Ring for manual manipulation, or (3) Spool for servo-based electrical control.

Printing Settings for Different Parts: To ensure compatibility with various multi-material 3D printers, Xstrings does not generate G-code for specific models but provides parameter settings for slicing software. We tested this method on three printers: Ultimaker S5, Ultimaker 3, and Bambu X1. The output includes three files: Main (PLA), Cable1 (Nylon), and Cable2 (Nylon). We printed these files with certain settings that avoid fusing during the printing job.

5 Interactive Demonstrations

Multi-Directional Tentacle Actuation: As mentioned in the previous section, we utilized the design tool to fabricate a controllable tentacle, which consists of 28 components and is printed at the maximum diagonal length (352 mm) of the 3D printing platform. This mechanism incorporates three cables that control the structure to bend left, coil at the tip, and bend right, respectively (as shown in Fig. 5 b1, b2, and b3). Due to the real-time responsiveness and accuracy of cable-driven tip control, the tip's coiling action enables better interaction with objects, such as grasping a pencil and wrapping a finger. The design of linked joints allows for precise control of the tip's spatial position, helping it to locate and drag objects more accurately. This case demonstrates how Xstrings utilizes cable-driven mechanisms in 3D-printed dynamic structure to achieve complex motions, providing next-level control and interactivity for FDM 3D printing.

Bio-Inspired Lizard Robot: Tasks such as exploring unknown areas, environmental modeling, and emergency assistance benefit from collaborative robot operations. While bio-inspired crawling robots excel at traversing complex terrains, their high manufacturing costs and complex production processes limit widespread application and mass production. To address this, we present a low-cost bio-inspired lizard robot based on the Xstrings cable-driven mechanism (Fig. 6). The robot mimics natural locomotion by alternately bending its body and lifting its back legs, creating a shuffling motion that utilizes surface friction and shifts its center of mass to

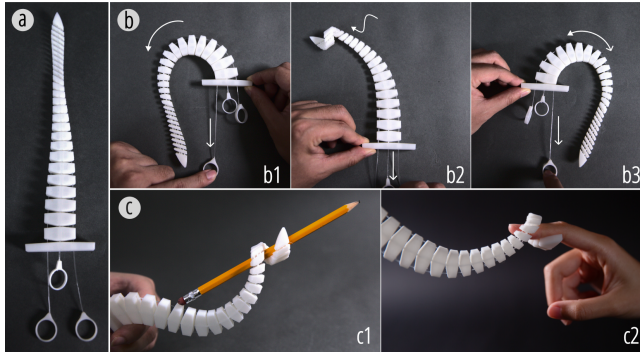


Figure 5: Controlled movements showing the tentacle's ability to perform bend and coil motions through different cable manipulation.

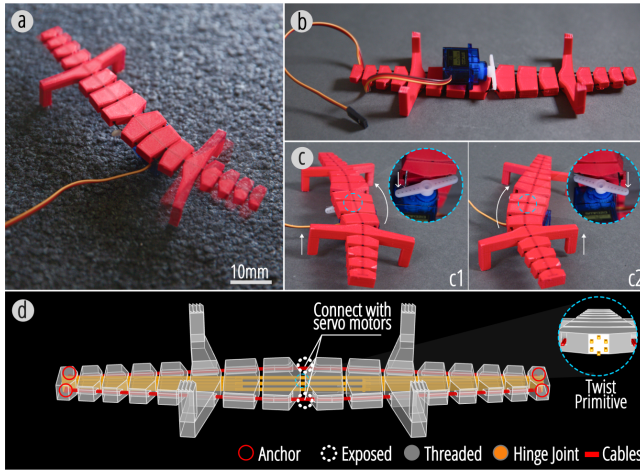


Figure 6: (a) Xstrings lizard robot demonstrating biomimetic locomotion.

propel forward. Precise cable actuation allows it to navigate narrow spaces and diverse terrains effectively. Using a print-in-place manufacturing method and economical components—\$1.90 for the motor and \$0.40 for printing material—this design offers a scalable solution for producing cable-driven mechanisms.

Claw with Embedded Actuation: In this section, we designed a cable-driven bird claw with an embedded actuation mechanism to demonstrate the Xstrings design and fabrication capabilities. This print-in-place structure features four fingers, each composed of five bending primitives connected by compliant joints, arranged in parallel and actuated by a single cable. The claw also includes a mechanical joint in the arm, enabling the elbow to bend and pull the cable, causing the claw to grip. This fully embedded mechanism allows users to pick up and transport objects by applying a single force to the elbow. Thanks to the ratchet design of the cable shape, the claw tightens securely during operation, ensuring a firm grip without releasing.

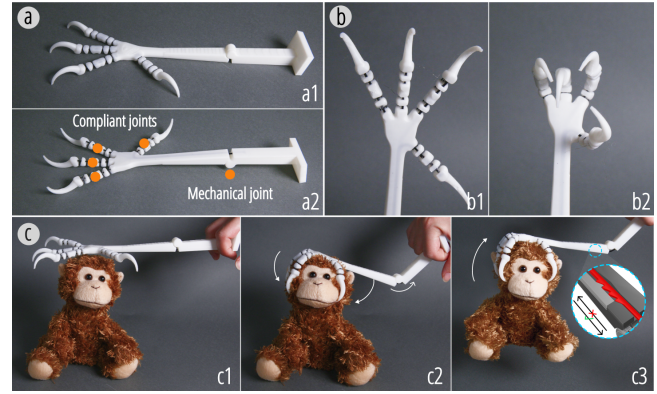


Figure 7: Design of the claw, featuring four fingers with compliant joints and an elbow with mechanical joint.

6 Conclusion

Our Xstrings method provides a novel approach to 3D printing cable-driven mechanisms, significantly simplifying the fabrication process by integrating cable-driven systems directly into the geometry of 3D-printed objects. Unlike traditional methods that require labor-intensive manual assembly, Xstrings automates the design and fabrication, supporting four types of cable-driven interactions—bend, twist, coil, and compress—using dual-material FDM 3D printing. It enables rapid prototyping and the production of complex dynamic structures without the need for additional assembly.

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